

Refereeing Gaps after Accounting for Playing Style

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Abstract

I compare recorded refereeing decisions for Barcelona in La Liga and Real Madrid in the Champions League, accounting for strength, opponent and playing style. A common-sample specification ladder separates changes in controls from changes in sample composition. Pooled Champions League comparisons improve precision relative to annual estimates, while power calculations state what remains detectable. The evidence describes differences in decisions. Without comparable incident opportunities and independent assessments of correctness, it does not establish favoritism.

1 Introduction

Possession-dominant teams face different opportunities to commit and suffer fouls. Raw differences in cards and fouls therefore do not identify favorable refereeing. I compare Barcelona with Real Madrid and non-elite La Liga clubs, and Real Madrid with other Champions League participants. The analysis reports adverse, favorable and imprecise estimates rather than selecting outcomes that support a claim about either club.

2 Data

The rebuilt sample contains 48,346 completed matches across the six competitions through 2025/26. 43 recorded possession pairs were invalid and were masked, not rescaled. 3,230 penalty-eligible matches would have been lost to the old low-penalty-rate rule. Finished seasons are not necessarily complete source coverage; data-audit-coverage.csv reports missing fixtures. 23 matches have conflicting ESPN and football-data scores, listed in data-audit-score-disagreements.csv. No source is declared correct by this audit. The internal Elo still uses ESPN results, so score errors can affect that strength control. I use the cached ESPN match summaries and football-data.co.uk league files. Penalties require a recorded regulation-end marker and goal totals that reconcile through extra time. This rule avoids selecting seasons using the number of penalties, but does not establish that every penalty event is correctly labeled. Table manifests report the outcome-specific estimation samples.

3 Empirical strategy

Tables use effects per 10 matches; figures label their own units. Positive means more favorable to the target club. Main comparison intervals are pointwise 95% season-block clustered intervals with small-sample t critical values. The two team rows from a match are not treated as independent games. Season blocks allow arbitrary dependence within a season but assume independence across

seasons. Models compare recorded decisions, not correct and incorrect calls. For Barcelona, L0 reports raw club indicators, L1 adds home, season, opponent and pre-match win-probability controls, and L2 adds splines in both teams’ season-to-date possession and shots from strictly earlier matches. L3 instead adds same-match possession and shots to L1; L4 adds both teams’ called fouls to L3 for card outcomes. The sample is held fixed across specifications within an outcome. Barcelona’s prior-match possession lies above the non-elite 95th percentile in 99.9% of its eligible matches. Only 1.3% lies inside Real Madrid’s central 90% range. The lagged-style models therefore depend on extrapolation, especially against non-elite clubs. L3 therefore uses same-match style instead of lagged style. It has more support but conditions on variables refereeing can change. The Champions League models hold fixed venue, season, stage, opponent, Elo expected-score bins, possession bins and both teams’ shots. The other-elite comparison excludes Barcelona, which has its own indicator. No domestic referee effects are transferred into this competition. Pooling all eligible seasons estimates a persistent average gap rather than a different effect in every small season. It adds information without inventing matches or treating domestic and UEFA officials as interchangeable. The MDE is calibrated under a noncentral-t approximation at 80% power and 5% two-sided size, using the estimated clustered SE. Equivalence results use explicit illustrative bounds and 90% intervals; failure to reject zero is not equivalence. Seasonal empirical-Bayes estimates retain the full covariance of jointly estimated season coefficients. Their intervals condition on estimated between-season heterogeneity and can understate uncertainty. Unpooled estimates, heterogeneity sensitivity, and leave-one-season-out fits remain available. Shrinkage intervals are not the headline evidence. The before/after-2018 comparison is descriptive. The timing also coincides with VAR and changes in squads and performance. It is not an identified treatment effect and is not presented as a causal event study.

4 Results

Table 1 reports the style-adjusted contrasts and their pointwise intervals. The browser report presents the complete specification ladder, exposure diagnostics, era contrasts and early-card comparisons. All reported numbers are generated from machine-readable manifests. Fouls committed have estimated possession elasticity 0.25, 95% CI [0.21, 0.28], with $p < 0.001$ against proportionality of one. Fouls suffered have estimated possession elasticity 0.24, 95% CI [0.21, 0.27], with $p < 0.001$ against proportionality of one.

4.1 Pooled Champions League comparisons

Table 2 pools every Champions League season with possession data and reports Real Madrid’s gaps per 10 matches, with intervals clustered by season and t criticals on 18 degrees of freedom; the sample is 218 Real Madrid matches in 19 seasons, with the same matches behind the raw, pre-match and style-adjusted columns. For net yellow cards, the style-adjusted gap is -2.39 cards per 10 matches (95 percent CI -5.66 to 0.87), equal in magnitude to 12 percent of the non-elite rate of 1.98 per match, and statistically indistinguishable from zero; the pre-match estimate is -1.91 , and the gap relative to the other elite clubs is -1.95 (95 percent CI -4.35 to 0.45 , $p = 0.105$). For net fouls, the style-adjusted gap is 6.40 fouls per 10 matches (95 percent CI -6.70 to 19.49), equal in magnitude to 5 percent of the non-elite rate of 13.22 per match, and statistically indistinguishable from zero; the pre-match estimate is 6.64, and the gap relative to the other elite clubs is 12.89 (95 percent CI 2.25 to 23.54, $p = 0.020$). For net penalties, the style-adjusted gap is -0.23 penalties per 10 matches (95 percent CI -1.44 to 0.98), equal in magnitude to 17 percent of the non-elite rate of 0.14 per match, and statistically indistinguishable from zero; the pre-match estimate is -0.01 ,

Table 1: Style-adjusted comparisons per ten matches

Target	Outcome	Gap [95 percent CI]	Holm p
Barcelona	Fewer yellow cards received	1.29 [-0.96, 3.55]	1.000
Barcelona	More opponent yellow cards	-0.75 [-2.37, 0.88]	1.000
Barcelona	Net yellow cards	0.55 [-1.85, 2.94]	1.000
Barcelona	Net fouls	3.26 [-7.98, 14.50]	1.000
Barcelona	Net red cards	-0.36 [-1.12, 0.41]	1.000
Barcelona	Net penalties	-0.10 [-0.95, 0.75]	1.000
Real Madrid	Net yellow cards	-1.95 [-4.35, 0.45]	0.941
Real Madrid	Net fouls	12.89 [2.25, 23.54]	0.203
Real Madrid	Net penalties	-0.64 [-1.61, 0.33]	1.000
Real Madrid	Net red cards	0.40 [-0.42, 1.21]	1.000
Real Madrid	Net yellow cards, given called fouls	-3.07 [-4.82, -1.33]	0.018

Barcelona is compared with Real Madrid in La Liga; Real Madrid is compared with other elite clubs excluding Barcelona in the Champions League. Positive values favor the target, including the sign-reversed own-yellow outcome. Intervals are pointwise season-block clustered intervals. Holm p-values adjust the family of eleven displayed endpoints. These are conditional associations, not estimates of incorrect decisions.

and the gap relative to the other elite clubs is -0.64 (95 percent CI -1.61 to 0.33 , $p = 0.180$). For net red cards, the style-adjusted gap is 0.22 red cards per 10 matches (95 percent CI -0.80 to 1.24), equal in magnitude to 23 percent of the non-elite rate of 0.10 per match, and statistically indistinguishable from zero; the pre-match estimate is 0.43 , and the gap relative to the other elite clubs is 0.40 (95 percent CI -0.42 to 1.21 , $p = 0.322$). Conditional on the fouls called, the net yellow-card gap is -2.98 cards per 10 matches (95 percent CI -5.43 to -0.53), statistically significant at the five percent level. Dropping the 8 Real Madrid matches that went to extra time moves the net yellow-card estimate to -2.34 and the net penalty estimate to -0.35 .

A null is informative only in proportion to what the sample could detect, so Figure 5 traces precision as seasons accumulate. Pooled over all seasons, the design detects, with 80 percent power, a net yellow-card gap of 4.60 cards per 10 matches (23 percent of the non-elite rate) and a net penalty gap of 1.71 penalties per 10 matches (124 percent); a single season resolves roughly 10.22 cards and 3.37 penalties at the same confidence. The 90 percent intervals rule out net yellow-card gaps beyond 5.09 cards and net penalty gaps beyond 1.23 penalties per 10 matches in either direction; these are equivalence margins against stated bounds, not evidence of no bias. Figure 6 shows that the pooled estimates do not rest on one season: leaving out each season in turn moves the net yellow-card gap across net yellow cards (-3.13 to -1.69 ; the refits straddle the five percent boundary), net fouls across net fouls (3.04 to 9.13 ; every refit covers zero), and net penalties across net penalties (-0.51 to 0.10 ; every refit covers zero). The season estimates themselves are more dispersed than sampling noise alone implies for net yellow cards (heterogeneity Q test $p = 0.03$, $I^2 = 43$ percent, between-season standard deviation 3.77 cards per 10 matches with an upper bound of 6.89), which is why the shrunk season estimates in the figure sit close to the pooled line; their intervals condition on the estimated between-season variance and are model-based rather than design-based.

5 Interpretation and limits

Possession is a proxy for opportunities, not a count of comparable tackles or penalty-area incidents. Fouls can occur off the ball. Yellow cards include dissent and time-wasting, so cards divided by fouls are not a binomial probability. Called fouls, final possession and shots can themselves respond to earlier decisions. The cached data lack foul severity, uncalled incidents, complete box touches and independently judged decision correctness. They also lack sufficiently complete referee identities for a modern referee-fixed-effect design. A residual gap cannot establish favoritism or rule out bias in a small number of decisive incidents. The earlier draft's causal language is not supported by this audit. Club ranks are descriptive, not permutation tests, because club labels are not exchangeable. The 2018 split is descriptive, not an identified effect of the reported payments. The stoppage variable measures the final-whistle clock rather than the announced minimum, and injury, substitution and time-wasting exposure is incomplete. I do not treat the earlier stoppage association as the clearest evidence of favoritism. A sharper correctness outcome would require an independently adjudicated set of comparable incidents, including no-calls, with foul severity, location, score state and time at risk. The present sources cannot supply that denominator. Official definitions distinguish misconduct from fouls and announced minimum added time from elapsed time: IFAB Law 12 and IFAB Law 7.

Table 2: Real Madrid’s Champions League Gaps per 10 Matches: Pooled Estimates and Detectable Effects

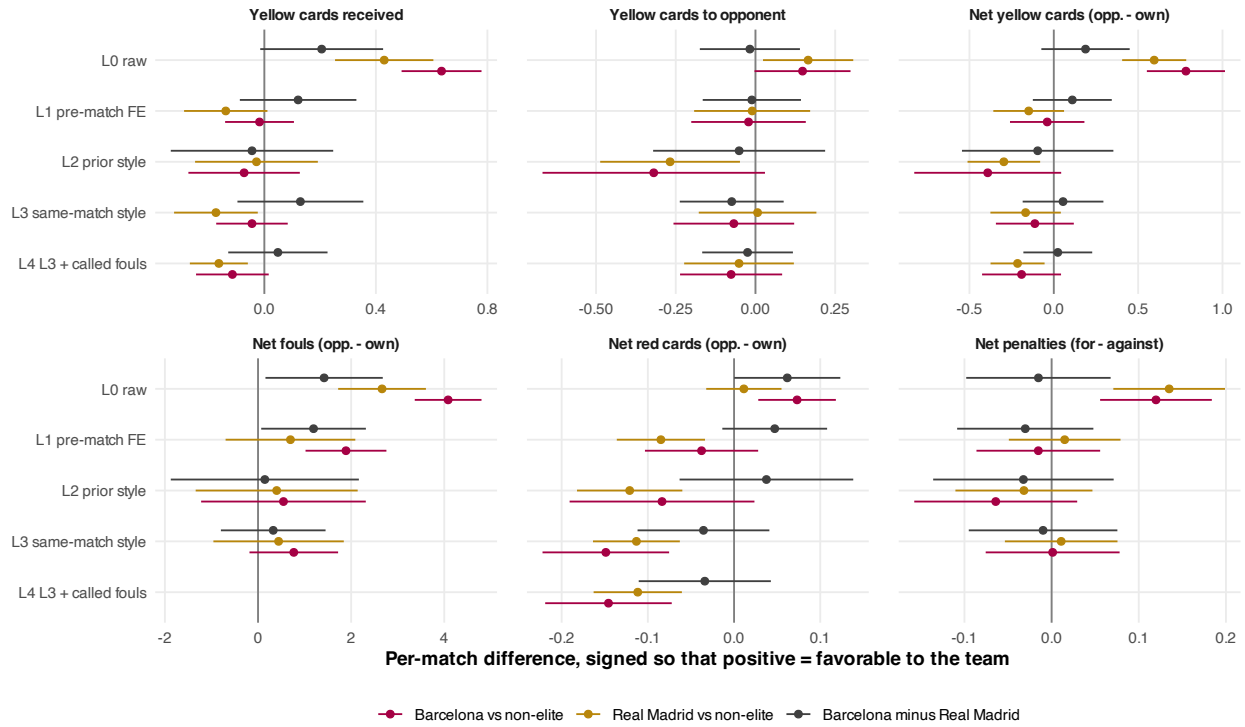
	vs non-elite clubs			vs other elite clubs			RM matches	MDE	Equiv.
	Raw	Pre-match	Style	Raw	Pre-match	Style	/ seasons	per 10	margin
Net yellow cards	2.63* [-0.24, 5.50]	-1.91 [-5.17, 1.34]	-2.39 [-5.66, 0.87]	-0.33 [-2.72, 2.06]	-1.63 [-4.07, 0.81]	-1.95 [-4.35, 0.45]	218 / 19	4.60	5.09
Net fouls	22.90*** [13.82, 31.98]	6.64 [-6.12, 19.39]	6.40 [-6.70, 19.49]	20.33*** [10.69, 29.98]	12.71** [2.26, 23.16]	12.89** [2.25, 23.54]	218 / 19	18.47	17.21
Net penalties	0.77 [-0.24, 1.78]	-0.01 [-1.22, 1.19]	-0.23 [-1.44, 0.98]	-0.34 [-1.33, 0.65]	-0.51 [-1.45, 0.43]	-0.64 [-1.61, 0.33]	218 / 19	1.71	1.23
Net red cards	0.46 [-0.35, 1.27]	0.43 [-0.63, 1.48]	0.22 [-0.80, 1.24]	0.39 [-0.39, 1.17]	0.46 [-0.39, 1.31]	0.40 [-0.42, 1.21]	218 / 19	1.44	1.07
Net yellow cards — fouls called	0.65 [-1.58, 2.87]	-2.51** [-4.96, -0.06]	-2.98** [-5.43, -0.53]	-2.11** [-3.98, -0.23]	-2.73*** [-4.54, -0.92]	-3.07*** [-4.82, -1.33]	218 / 19	3.45	5.00

Notes: Champions League group/league-phase and knockout team-matches with valid possession shares; Real Madrid plays in 2007/08–2025/26. Coefficients are per-match gaps multiplied by 10, positive when favorable to Real Madrid; a match that went to extra time counts once. Raw controls for venue only; pre-match adds season, stage, opponent and Elo expected-score ventile fixed effects; style adds 2.5-point own-possession bins and both teams’ shots, a conditional association rather than a bias estimate. “Net yellow cards | fouls called” adds own and opponent fouls as controls. Brackets: 95 percent confidence intervals from standard errors clustered by season with t criticals on the number of seasons minus one. MDE: minimum detectable effect at 80 percent power and 5 percent two-sided size for the style column vs non-elite clubs. Equiv. margin: the smallest symmetric bound within which the style estimate is TOST-equivalent at 5 percent (the wider end of the 90 percent interval). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ against the season-block t distribution. Source: ESPN. Run 20261003-175015.

Figure 1: Barcelona: raw and adjusted differences

Barcelona's La Liga decisions across a common-sample specification ladder

Same matches; L2 and L3 are alternative style controls. 95% season-block CIs with $t(G-1)$ critical values.

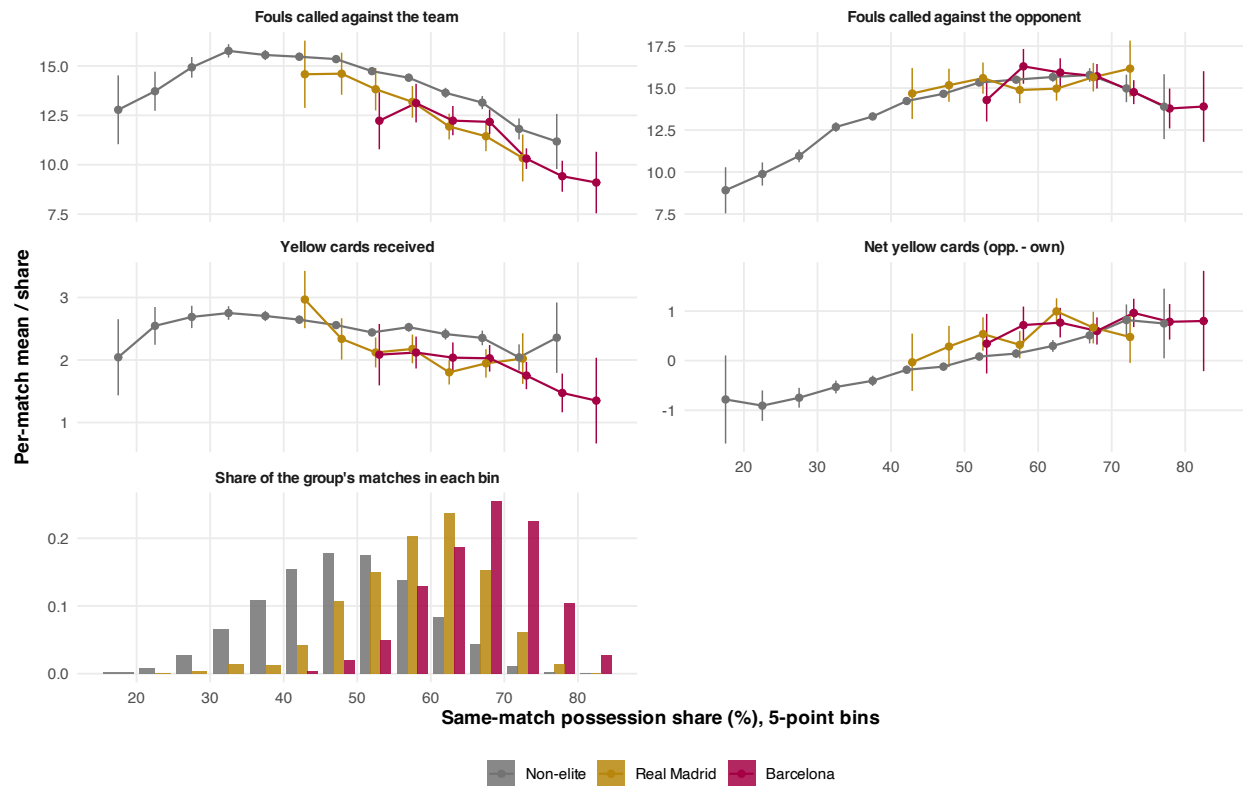


Common sample: 7,255 La Liga matches (14,510 team-match rows, 21 seasons, 714 Barcelona rows); penalties on 7,236 matches with eligible event logs. L2 uses season-to-date possession and shots of both teams over prior matches only. L3 uses same-match style instead; these controls respond to decisions. Coefficients are conditional associations, not estimates of favoritism. Sources: ESPN, football-data.co.uk.

Figure 2: Possession and the observed decision rates

Fouls and cards by possession: Barcelona against teams at the same possession

Bins with at least 15 matches per group; bars show where each group's matches sit. Poisson elasticity of fouls against the team with respect to opponent possession: 0.25 [0.21, 0.28]; fouls against the opponent w.r.t. own possession: 0.24 [0.21, 0.27]

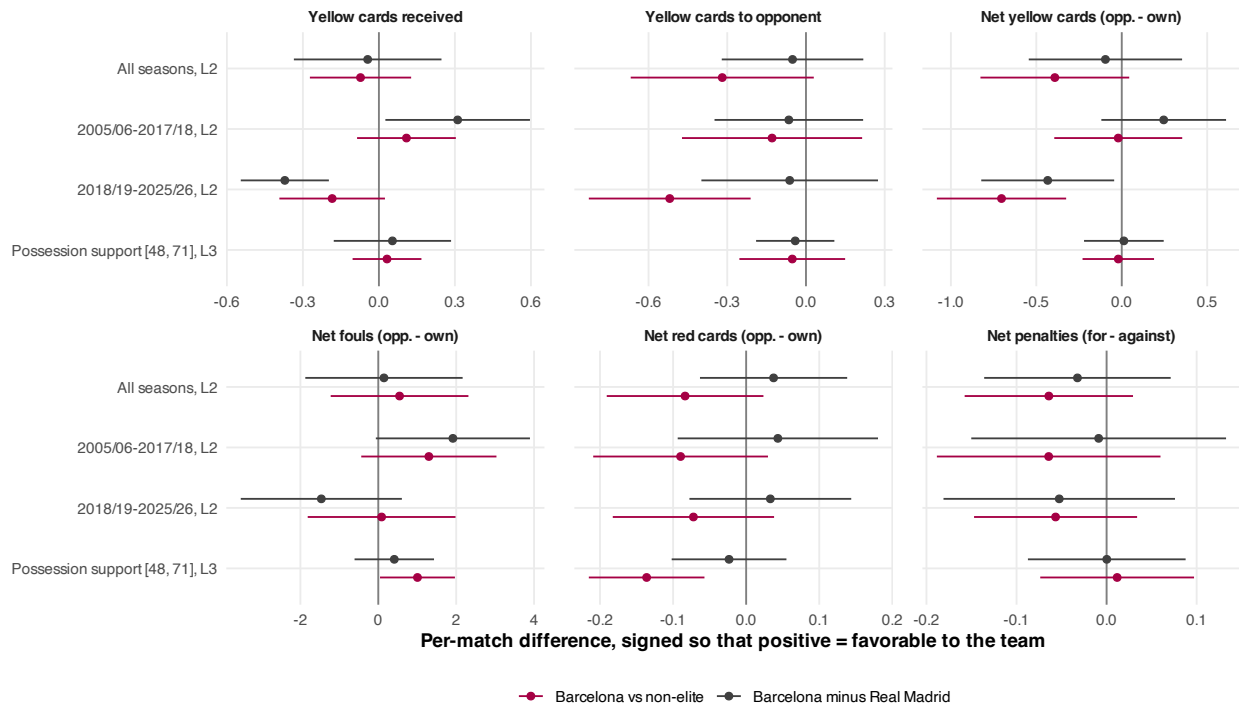


Means with unclustered 95% intervals, descriptive only; same-match possession responds to decisions. Common sample as in the ladder figure. Elasticities from Poisson regressions with season, opponent and win-probability FE; season-cluster CIs. Sources: ESPN, football-data.co.uk.

Figure 3: Barcelona: eras and overlap

Era split and possession common support: Barcelona's conditional gaps

Eras use prior-match style controls. The overlap sample uses same-match style.
Both comparisons are descriptive; trimming changes the population.

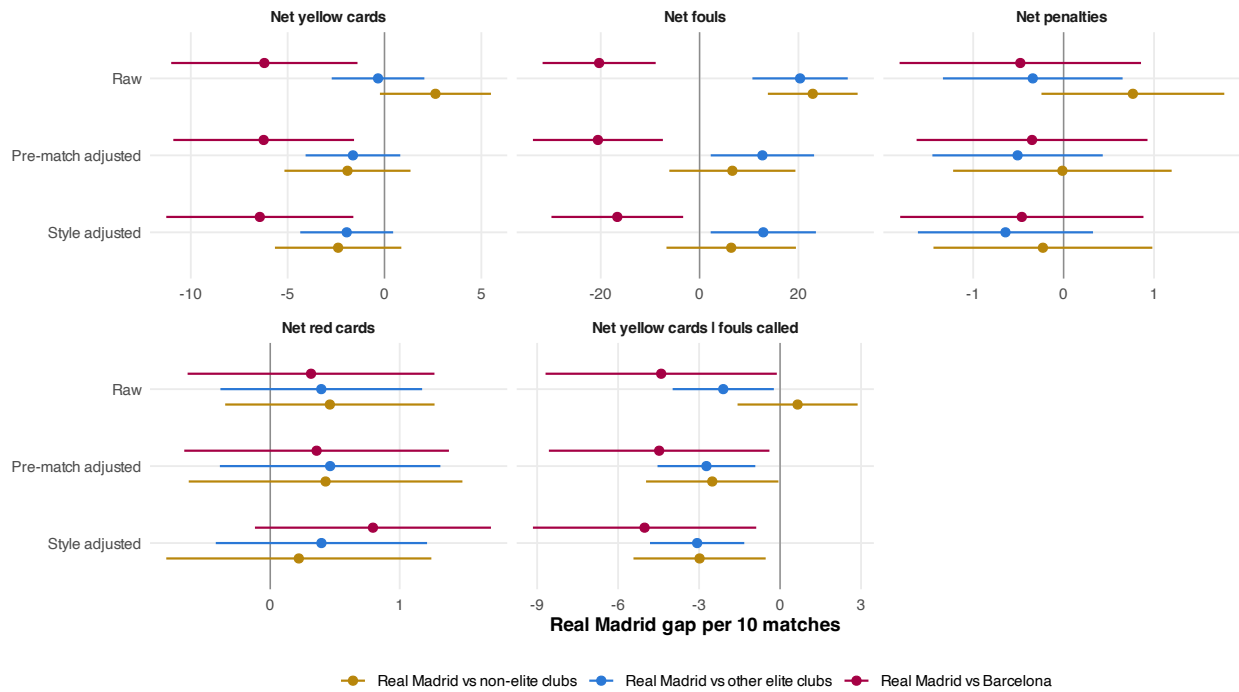


2018 separates the reported payment period and first VAR season; it does not identify a payment effect.
Possession support drops 219 of 714 Barcelona rows (31%) and the low-possession side of most matches.
95% season-block CIs with t(G-1) critical values. Sources: ESPN, football-data.co.uk.

Figure 4: Madrid: pooled Champions League comparisons

Real Madrid in the Champions League: pooled refereeing gaps

Same sample across the three columns: matches with valid possession and shots. 218 Real Madrid matches in 19 seasons (2007/08 to 2025/26).

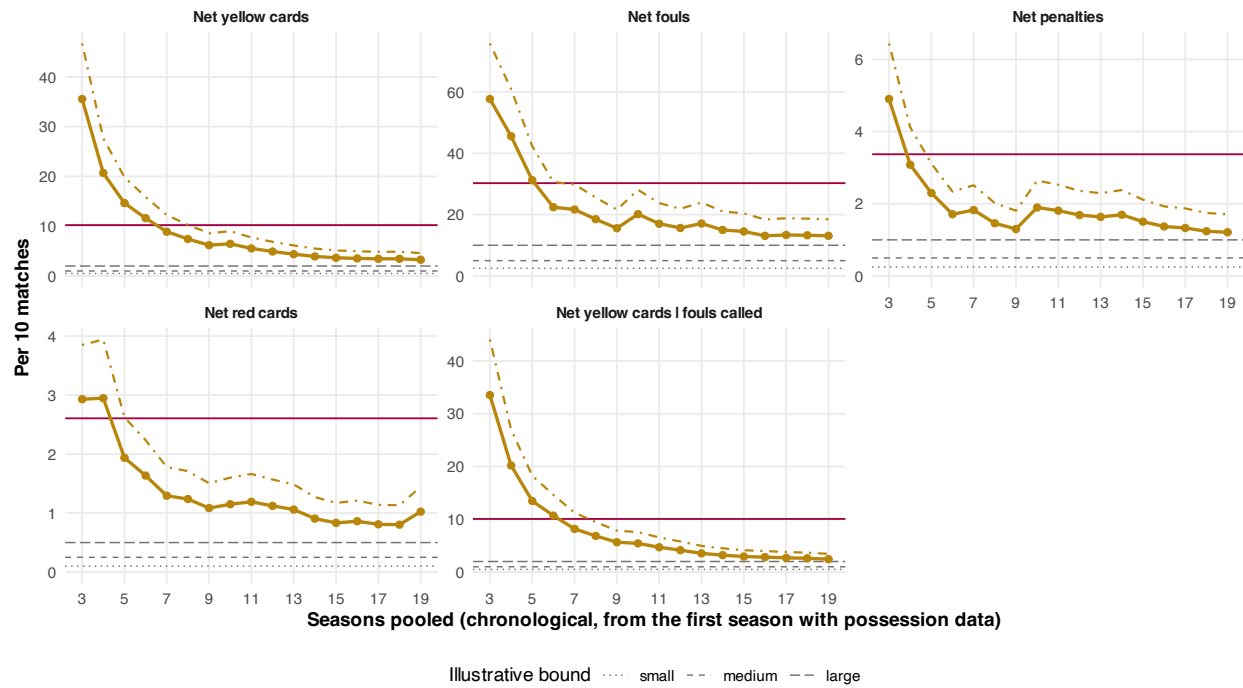


Raw: venue only. Pre-match: season, stage, opponent and Elo expected-score ventile fixed effects. Style: adds own-possession bins and both teams' shots. These same-match controls respond to decisions; the style column is descriptive. 'fouls called' conditions net yellow cards on own and opponent fouls. Intervals: 95% season-block cluster (CR1) with t criticals on seasons minus one. Positive = favorable to Real Madrid. Unit: team-match; matches with extra time count once. Source: ESPN.

Figure 5: Madrid: what the sample can detect

What the Champions League sample can resolve for Real Madrid

Gold solid: 95% CI half-width of the pooled style-adjusted gap vs non-elite clubs. Gold dot-dash: minimum detectable effect (80% power, 5% two-sided). Red: median single-season 95% CI half-width. Grey lines: illustrative equivalence bounds, not claims about relevance.

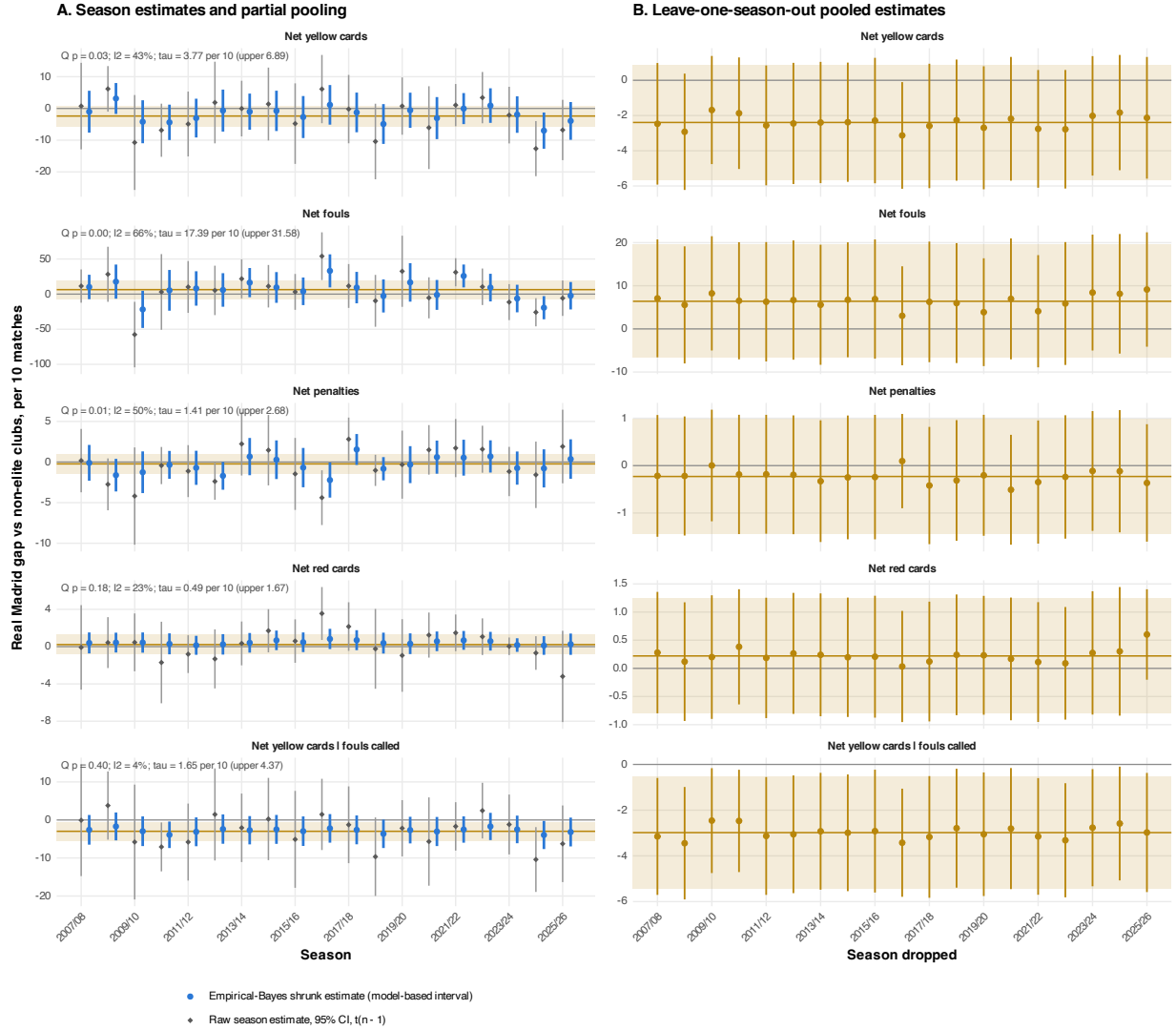


Each point refits the pooled model on all matches through that season; no matches are added or reweighted. Precision gains come from more seasons and from larger t -criticals relaxing as clusters grow. Intervals: 95% season-block cluster (CR1) with t -criticals on seasons minus one. Positive = favorable to Real Madrid. Unit: team-match; matches with extra time count once. Source: ESPN.

Figure 6: Madrid: seasonal uncertainty and stability

Does the pooled Real Madrid estimate depend on one season?

Gold line and band: pooled style-adjusted estimate with its 95% season-block CI. Shrinkage is empirical Bayes at the REML tau; its intervals condition on the estimated tau and understate uncertainty when tau is near zero. Q tests heterogeneity; I² summarizes dispersion beyond sampling noise.

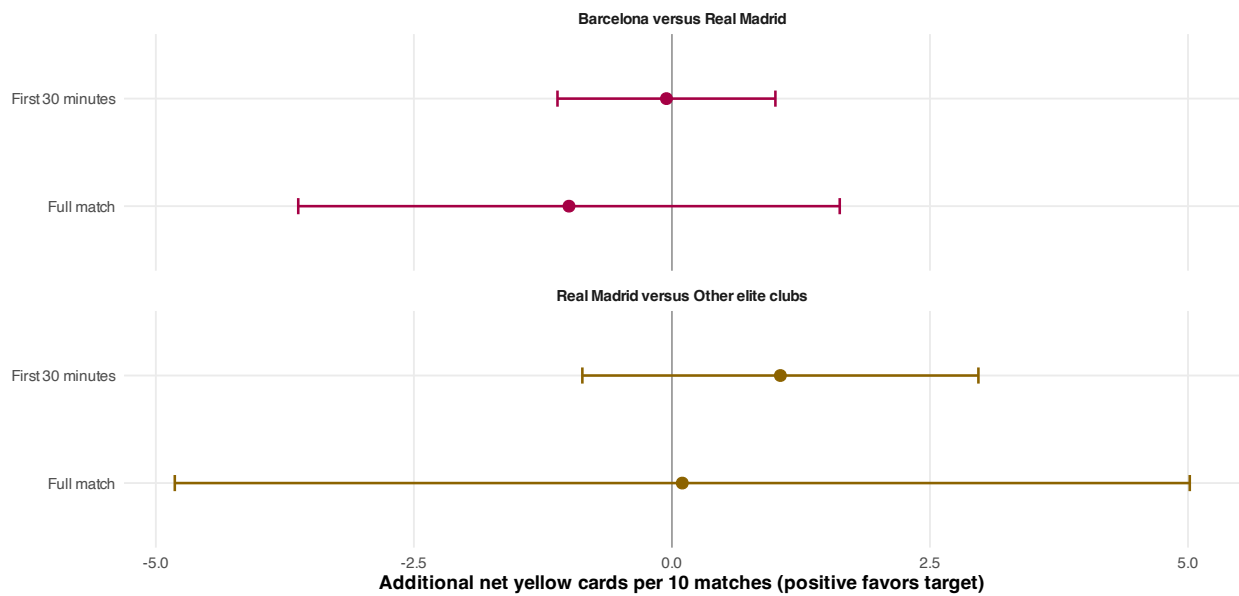


Left: season coefficients from one model with season x Real Madrid interactions, match-pair clustered SEs, t(n - 1) with n the season's Real Madrid matches. Right: refits of the pooled model without each season. Intervals: 95% season-block cluster (CR1) with t criticals on seasons minus one. Positive = favorable to Real Madrid. Unit: team-match; matches with extra time count once. Source: ESPN.

Figure 7: Early-card diagnostic

Does the card gap appear early in the match?

Same matches in each panel; prior-match style, not final possession, is held fixed.



Bars: 95% season-block clustered intervals. Event totals must match both box scores; extra time excluded.
First 30 minutes and full-match counts have different exposure. Timing does not establish whether calls were correct.